

# RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

## 6. RF Design Practices

10 hrs · asked in **all 24 papers** · 3 syllabus sub-topics · **426 marks, 22.4% of the archive, the heaviest chapter** → one filter question and one amplifier numerical in almost every paper.

### READ THIS FIRST

Chapter 6 is worth about 18 marks a paper and it is **two fixed questions**:

1. **A filter question** (20 of 24 papers): “why the insertion loss method”, “how is a Butterworth / Chebyshev LPF prototyped”, “design steps”, then “sketch it in microstrip”. → §6.1 to §6.3. The theory is one page; the marks are in the **procedure** and the **layout sketch**.
2. **An amplifier numerical** (22 papers, 10 to 18 marks): a transistor’s four S-parameters, “check stability and compare maximum gain for bilateral and unilateral modes”. → theory §6.4, every PYQ set solved in the companion. The papers **attach the formula sheet**; what they mark is the order of the steps and the arithmetic.

Two things worth knowing before the paper starts:

- Of the 13 distinct S-parameter sets in the archive, **three are not unconditionally stable** (the  $0.894\angle -60.6^\circ$  set in 4 papers, 76 Bh, 73 Ma). For those a “bilateral maximum gain” does not exist and the answer must say so. The companion shows what to write instead.
- Oscillator and mixer theory are asked in only 2 papers each, as short notes (§6.5, §6.6).

### 6.1 Microwave Filters and the Insertion Loss Method

**Why are microwave filters designed using the insertion loss method? How is a Butterworth / Chebyshev LPF prototyped?** 4 5 2+6 3+5 6+2 8 10

**TOP 20/24** 70 Bh, 70 Ma, 71 Bh, 72 Ash, 72 Ma, 73 Bh, 73 Ma, 74 Bh, 74 Ma, 75 Bh, 76 Bh, 77 Ch, 78 Ch, 79 Bh, 80 Ba, 80 Bh, 81 Ba, 81 Bh, 82 Ba, 82 Bh · **the most repeated ask in the chapter**

#### Filters, and the two design methods

- **Filter**: two-port that passes a frequency band and attenuates the rest: LPF, HPF, BPF, BSF.
- **Image parameter method** (1930s): cascade simple sections with the right cut-off and attenuation.
  - Cannot specify the **whole** response; needs repeated iteration.
- **Insertion loss method (ILM)**: network synthesis from a specified **power loss ratio** over all frequencies.

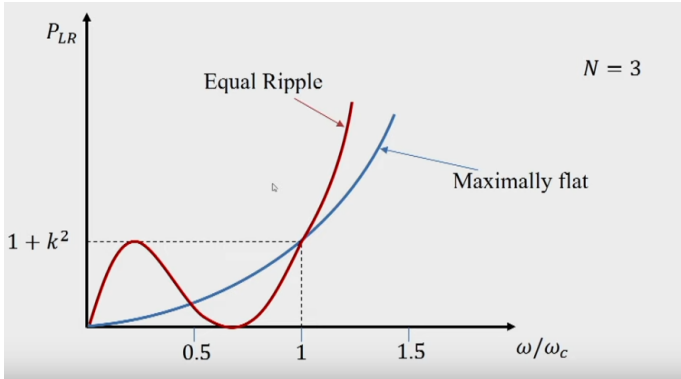
#### Power loss ratio

$$P_{LR} = \frac{P_{\text{available}}}{P_{\text{load}}} = \frac{1}{1 - |\Gamma(\omega)|^2} = 1 + \frac{M(\omega^2)}{N(\omega^2)}$$

- $|\Gamma|^2$  is even in  $\omega$ , so  $M, N$  are polynomials in  $\omega^2$ .
- Insertion loss  $IL = 10 \log P_{LR}$  dB; matched both ends:  
 $P_{LR} = 1/|S_{21}|^2$ .

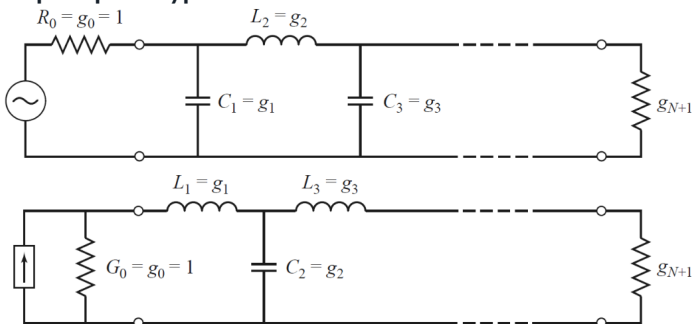
### Why the insertion loss method (the 2 to 3 mark answer)

- Response is **completely specified** and met exactly, no trial and error.
- Explicit trade-offs: **binomial** for minimum passband loss, **Chebyshev** for sharpest cut-off, **linear phase** for low distortion.
- Order  $N$  follows from the stopband requirement.
- Starts from **normalised low-pass prototypes** ( $R_0 = 1 \Omega$ ,  $\omega_c = 1$  rad/s), so one table serves every filter after scaling.
- Designs improve systematically with  $N$ ; CAD tools implement it directly.



$N = 3$ : equal-ripple is sharper, maximally flat is smoother. Gangaju Ch-6 deck, slide 11.

### Low-pass prototype and element values



The two dual ladder forms,  $g_0 = 1$  source,  $g_{N+1}$  load. Deck slide 7 (Pozar p.403).

- Butterworth** ( $k = 1$ ):  $g_k = 2 \sin \frac{(2k-1)\pi}{2N}$ ,  $g_{N+1} = 1$ .
- Chebyshev**, ripple  $L_r$  dB:  $\beta = \ln \coth \frac{L_r}{17.37}$ ,  $\gamma = \sinh \frac{\beta}{2N}$ ,  
 $a_k = \sin \frac{(2k-1)\pi}{2N}$ ,  $b_k = \gamma^2 + \sin^2 \frac{k\pi}{N}$ ;  $g_1 = \frac{2a_1}{\gamma}$ ,  
 $g_k = \frac{4a_{k-1}a_k}{b_{k-1}g_{k-1}}$ ;  $g_{N+1} = 1$  for odd  $N$ ,  $\coth^2 \frac{\beta}{4}$  for even  $N$ .
- Order** (Butterworth): smallest  $N$  with  $10 \log [1 + (\omega/\omega_c)^{2N}] \geq$  required stopband loss.

### The three standard responses

- Maximally flat (Butterworth, binomial):**  
 $P_{LR} = 1 + k^2(\omega/\omega_c)^{2N}$ .  
 - Band-edge loss  $1 + k^2$ ;  $k = 1$  gives  $-3$  dB.  
 - Beyond cut-off rises  $20N$  dB/decade; flattest possible passband.
- Equal ripple (Chebyshev):**  $P_{LR} = 1 + k^2 T_N^2(\omega/\omega_c)$ .  
 - Ripple of amplitude  $1 + k^2$  in the passband.  
 - Also  $20N$  dB/decade, but  $\frac{2^{2N}}{4}$  times more attenuation than Butterworth well beyond cut-off.
- Linear phase:**  $\phi = A\omega[1 + p(\omega/\omega_c)^{2N}]$ , maximally flat group delay; poorer selectivity.

### Tables (computed in filt.py, equal to Pozar's)

| $N$                         | $g_1$  | $g_2$  | $g_3$  | $g_4$  | $g_5$  | $g_6$  |
|-----------------------------|--------|--------|--------|--------|--------|--------|
| <b>Maximally flat</b>       |        |        |        |        |        |        |
| 1                           | 2.0000 | 1.0000 |        |        |        |        |
| 2                           | 1.4142 | 1.4142 | 1.0000 |        |        |        |
| 3                           | 1.0000 | 2.0000 | 1.0000 | 1.0000 |        |        |
| 4                           | 0.7654 | 1.8478 | 1.8478 | 0.7654 | 1.0000 |        |
| 5                           | 0.6180 | 1.6180 | 2.0000 | 1.6180 | 0.6180 | 1.0000 |
| <b>Equal ripple, 0.5 dB</b> |        |        |        |        |        |        |
| 1                           | 0.6987 | 1.0000 |        |        |        |        |
| 2                           | 1.4029 | 0.7071 | 1.9841 |        |        |        |
| 3                           | 1.5963 | 1.0967 | 1.5963 | 1.0000 |        |        |
| 4                           | 1.6704 | 1.1925 | 2.3662 | 0.8419 | 1.9841 |        |
| 5                           | 1.7058 | 1.2296 | 2.5409 | 1.2296 | 1.7058 | 1.0000 |
| <b>Equal ripple, 3.0 dB</b> |        |        |        |        |        |        |
| 1                           | 1.9954 | 1.0000 |        |        |        |        |
| 2                           | 3.1014 | 0.5339 | 5.8095 |        |        |        |
| 3                           | 3.3489 | 0.7117 | 3.3489 | 1.0000 |        |        |
| 4                           | 3.4391 | 0.7483 | 4.3473 | 0.5920 | 5.8095 |        |
| 5                           | 3.4815 | 0.7619 | 4.5378 | 0.7619 | 3.4815 | 1.0000 |

- Deck example: 20 dB at 4 GHz for  $f_c = 2.5$  GHz. The deck (via Pozar's chart) takes  $N = 6$  (24.5 dB); the formula shows  $N = 5$  already gives 20.4 dB.  $N = 6$  is safe, just not minimal. verified/added

## Why are Butterworth and Chebyshev common to prototype two-port filters? 4

70 Bh, 70 Ma

- Both power loss ratios are **polynomials** in  $\omega^2$  that give a **realisable lossless LC ladder** with closed-form element values.
- They sit at the two ends of the useful trade-off: **flattest passband** (Butterworth) vs **sharpest cut-off** for a given  $N$  (Chebyshev).
- Tabulated, normalised values mean a designer only **scales**; the same table gives LPF, HPF, BPF and BSF by transformation (§6.2).
- Ladders map directly onto transmission-line sections, so both are practical in microstrip (§6.3).

## 6.2 Filter Design Steps: Scaling and Conversion

### Explain the steps of microwave filter design; how is a LPF prototype converted to other filters? 5 6+4

7+3 3+8+3 10

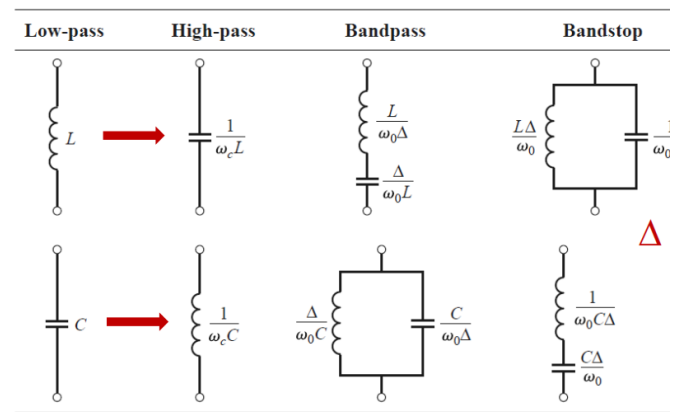
same papers as §6.1 · 71 Bh, 72 Ma, 74 Bh, 80 Bh, 81 Ba, 82 Ba, 82 Bh ask the steps outright

#### The procedure (write it as four boxes)

- Specifications:** type, cut-off  $f_c$  (or band  $f_1$  to  $f_2$ ), passband loss or ripple, stopband attenuation at a stated frequency,  $R_0$ .
- Low-pass prototype:** choose response, find  $N$ , read  $g_1$  to  $g_{N+1}$  ( $R_0 = 1$ ,  $\omega_c = 1$ ).
- Scaling and conversion:** impedance-scale by  $R_0$ , frequency-scale to  $\omega_c$ , and transform LP  $\rightarrow$  HP / BP / BS.
- Implementation:** lumped (below about 5 GHz) or distributed microstrip: stepped impedance, Richards' stubs, Kuroda identities (§6.3).

#### Impedance and frequency scaling (LPF)

$$L'_k = \frac{R_0 L_k}{\omega_c}, \quad C'_k = \frac{C_k}{R_0 \omega_c}, \quad R'_s = R'_L = R_0$$



Element transformations,  $\Delta = (\omega_2 - \omega_1)/\omega_0$ ,  $\omega_0 = \sqrt{\omega_1 \omega_2}$ . Deck slide 19 (Pozar p.414).

#### Conversion rules

- LPF  $\rightarrow$  HPF**,  $\omega \rightarrow -\omega_c/\omega$ : series L becomes series  $C = \frac{1}{R_0 \omega_c L_k}$ ; shunt C becomes shunt  $L = \frac{R_0}{\omega_c C_k}$ .
- LPF  $\rightarrow$  BPF**: series L  $\rightarrow$  series LC ( $L_k R_0 / \omega_0 \Delta$ ,  $\Delta / \omega_0 L_k R_0$ ); shunt C  $\rightarrow$  shunt LC ( $\Delta R_0 / \omega_0 C_k$ ,  $C_k / \omega_0 \Delta R_0$ ).
- LPF  $\rightarrow$  BSF**: the reverse pairing (series arms become parallel LC, shunt arms series LC).

#### Worked scaling check: Butterworth $N = 3$ , $f_c = 2.5$ GHz, $R_0 = 50 \Omega$

| Prototype          | LPF element                                                            | HPF element                                                    | Check (ABCD in <code>filt.py</code> )                   |
|--------------------|------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| $g_1 = 1$ (shunt)  | $C_1 = \frac{1}{50 \times 2\pi \times 2.5\text{e9}} = 1.27 \text{ pF}$ | shunt $L = \frac{50}{\omega_c} = 3.18 \text{ nH}$              | $ S_{21}  = -3.01 \text{ dB at } 2.5 \text{ GHz, both}$ |
| $g_2 = 2$ (series) | $L_2 = \frac{50 \times 2}{\omega_c} = 6.37 \text{ nH}$                 | series $C = \frac{1}{50 \omega_c \times 2} = 0.637 \text{ pF}$ |                                                         |
| $g_3 = 1$ (shunt)  | $C_3 = 1.27 \text{ pF}$                                                | shunt $L = 3.18 \text{ nH}$                                    |                                                         |

### 6.3 Microstrip Implementation

Implement the filter using microstrip: LPF  $\pi$  / double  $\pi$ , double-pad T-type, double-section shunt-arm or series-arm, HPF, single-section series-arm BPF 2 3 3+5 5+3 7+3 8+2 10

TOP 13/24 72 Ash, 73 Ma, 74 Bh, 74 Ma, 76 Bh, 78 Ch, 79 Bh, 80 Ba, 80 Bh, 81 Ba, 81 Bh, 82 Ba, 82 Bh · the sketch carries the marks; label every section

#### Why microstrip, and the two methods

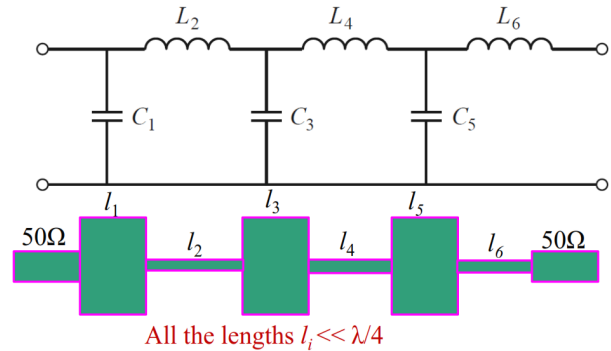
- Lumped L, C work to about 5 GHz; above that their size and spacing are no longer small in wavelengths.
- Microstrip: planar, cheap, light, easy to mount active devices; but lossier, lower power, radiates and couples.
- **Stepped impedance** (Hi-Z / Low-Z): alternate very high and very low impedance short lines.
- **Stubs** (Richards' transformation): inductor  $\rightarrow$  short-circuited stub, capacitor  $\rightarrow$  open-circuited stub, each  $\lambda/8$  long at  $f_c$ ; Kuroda identities separate series stubs.

#### Why a short line is an L or a C

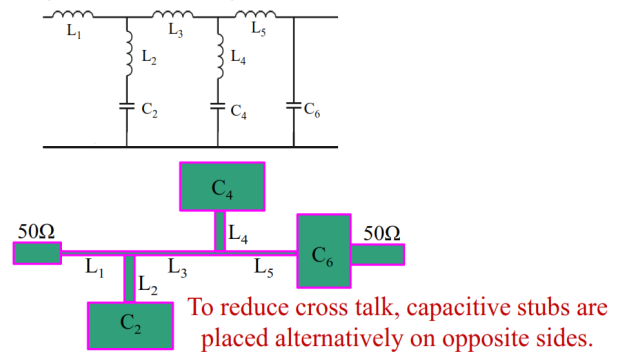
- T-model of a line length  $l$ : series  $X/2 = Z_0 \tan(\beta l/2)$ , shunt  $B = \sin(\beta l)/Z_0$ .
- **High  $Z_0$ , short**:  $X \approx Z_h \beta l$ ,  $B \approx 0 \rightarrow$  a **series inductor**.
- **Low  $Z_0$ , short**:  $B \approx \beta l/Z_l$ ,  $X \approx 0 \rightarrow$  a **shunt capacitor**.
- Design lengths:

$$\beta l = \frac{g_k R_0}{Z_h} \text{ (inductor)}, \quad \beta l = \frac{g_k Z_l}{R_0} \text{ (capacitor)}$$

Keep  $\beta l < 45^\circ$ ; typically  $Z_h \approx 100$  to  $120 \Omega$ ,  $Z_l \approx 20 \Omega$ .



Stepped-impedance LPF and its layout. Deck slide 26.



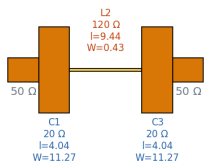
Elliptic LPF with capacitive stubs alternating sides to cut cross-talk. Deck slide 27.

- **Checked:** `filt.py` reproduces the deck's example table ( $N = 6$ ,  $Z_h = 120 \Omega$ ,  $Z_l = 20 \Omega$ ,  $\epsilon_r = 4.2$ ,  $h = 1.58$  mm):  $\beta l = 11.8, 33.8, 44.3, 46.1, 32.4, 12.3^\circ$ ,  $W = 11.3$  and  $0.43$  mm.

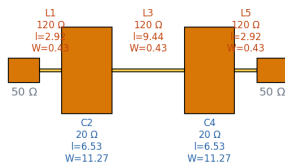
#### Vocabulary the papers use

- **Pad**: a wide, low- $Z$  section = a **shunt capacitor**.
- **Series arm**: an element in the signal path (series L in an LPF, series C in an HPF). **Shunt arm**: an element to ground (shunt C as a pad or open stub; shunt L as a shorted stub).
- $\pi$ -**section**: shunt, series, shunt (C-L-C for a LPF). **T-section**: series, shunt, series (L-C-L).
- **Double / triple**: two or three such sections cascaded.

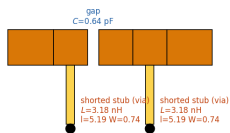
The layouts below are drawn to scale by `filt.py` from computed dimensions for one assumed specification, which is what a paper that says "illustrate an example" expects: Butterworth,  $f_c = 2.5$  GHz,  $R_0 = 50 \Omega$ ,  $Z_h = 120 \Omega$ ,  $Z_l = 20 \Omega$ ,  $\epsilon_r = 4.2$ ,  $h = 1.58$  mm ( $W_{50} = 3.13$  mm).

**LPF  $\pi$ -section,  $N = 3$  (79 Bh, 73 Ma, 74 Ma, 72 Ash)** $\pi$ -section LPF (C-L-C),  $N = 3$  (mm)

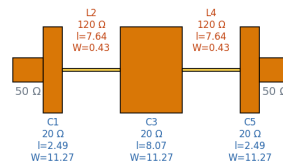
- $g = 1, 2, 1$ : pads  $\beta l = 22.9^\circ$  (4.04 mm), line  $47.7^\circ$  (9.44 mm).

**Double-pad T-type,  $N = 5$  (81 Ba; series-arm, 78 Ch)**Double-pad T-type LPF (L-C-L-C-L),  $N = 5$ : two low- $Z$  pads (mm)

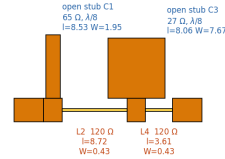
- L-C-L-C-L: the **series arms** are the three high- $Z$  lines, the two **pads** the shunt capacitors. A “double-section series-arm” filter is this layout.

**HPF  $\pi$ -section (74 Bh, 80 Bh, 82 Ba)**HPF  $\pi$ -section (shunt L - series C - shunt L),  $N = 3$ ,  $f_c = 2.5$  GHz (mm)

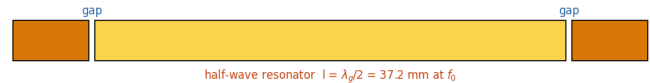
- From the C-L-C prototype: shunt  $L = 3.18$  nH, series  $C = 0.637$  pF.
- Series capacitor = a **gap** (or interdigital) in the strip; shunt inductor = a **short-circuited stub** through a via:  $Z_s \tan \beta l = \omega_c L$  gives 5.19 mm of 100  $\Omega$  line.
- **Triple-pad series-arm HPF** (82 Ba): three pads separated by two gaps (the series capacitors) with shorted stubs from the pads.
- 80 Bh's “second-order  $\pi$ ”: one series gap between two stubbed pads.

**Double  $\pi$ -section,  $N = 5$  (76 Bh)**Double  $\pi$ -section LPF (C-L-C-L-C),  $N = 5$  (dimensions in mm)

- $g = 0.618, 1.618, 2, 1.618, 0.618$ : three pads, two high- $Z$  lines.

**Double-section shunt-arm LPF,  $N = 4$  (80 Ba)**Shunt-arm LPF: open stubs for shunt C, high- $Z$  lines for series L,  $N = 4$  (mm)

- Shunt capacitors as **open stubs**,  $\lambda/8$  at  $f_c$  with  $Z = R_0/g_k$  (Richards): 65 and 27  $\Omega$ ; series inductors as high- $Z$  lines.

**Single-section series-arm BPF (81 Bh)**Single-section series-arm BPF: gap-coupled half-wave resonator,  $f_0 = 2.25$  GHz

- The series LC arm of a BPF is a **half-wave resonator** coupled by a gap at each end: resonant at  $f_0$  (series-resonant, passes), capacitively coupled (blocks DC and low frequencies).
- $l = \lambda_g/2$  at  $f_0$ : 37.2 mm on 50  $\Omega$  at 2.25 GHz.
- More sections (more resonators) sharpen the skirts.

## 6.4 RF Amplifier Theory: Gain, Stability and Design Flow

Draw the design flowchart of a microwave amplifier; stability analysis; stability synthesis on the Smith chart; amplifier vs oscillator 4 5 3+8 5+5 6+4 8 10

TOP 10/24 71 Bh, 71 Ma, 73 Bh, 74 Bh, 75 Bh, 76 Bh, 78 Ch, 81 Ba, 81 Bh, 82 Ba · the theory half of the amplifier questions

### Reflection coefficients seen by the device

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} = \frac{S_{11} - \Delta\Gamma_L}{1 - S_{22}\Gamma_L}$$

$$\Gamma_{out} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S} = \frac{S_{22} - \Delta\Gamma_S}{1 - S_{11}\Gamma_S}$$

$\Delta = S_{11}S_{22} - S_{12}S_{21}$ . From  $b_2 = S_{21}a_1 + S_{22}\Gamma_L b_2$  (deck slides 33 to 34).

### Three power gains

- **Transducer**  $G_T = P_L/P_{avs}$  (the one the papers use).
- **Available**  $G_A = P_{avn}/P_{avs}$ ; **operating**  $G_P = P_L/P_{in}$ .

$$G_T = \frac{1 - |\Gamma_S|^2}{|1 - \Gamma_{in}\Gamma_S|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

- **Unilateral** ( $S_{12} \approx 0$ ):  $G_{TU} = G_S G_0 G_L$  with  $G_0 = |S_{21}|^2$ ; maximum at  $\Gamma_S = S_{11}^*$ ,  $\Gamma_L = S_{22}^*$ :

$$G_{TU,max} = \frac{1}{1 - |S_{11}|^2} |S_{21}|^2 \frac{1}{1 - |S_{22}|^2}$$

- **Unilateral figure of merit**  $M = \frac{|S_{11}||S_{12}||S_{21}||S_{22}|}{(1 - |S_{11}|^2)(1 - |S_{22}|^2)}$ ; error bound  $\frac{1}{(1+M)^2} < \frac{G_T}{G_{TU}} < \frac{1}{(1-M)^2}$ . Unilateral is acceptable if the bound is a few tenths of a dB.
- **Bilateral maximum** (simultaneous conjugate match,  $\Gamma_S = \Gamma_{in}^*$ ,  $\Gamma_L = \Gamma_{out}^*$ ), only if unconditionally stable:

$$\Gamma_S = \frac{B_1 - \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}, \quad \Gamma_L = \frac{B_2 - \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2, C_1 = S_{11} - \Delta S_{22}^* \text{ (and } 1 \leftrightarrow 2\text{)}. \text{ Then } G_{T,max} = \frac{|S_{21}|}{|S_{12}|} (K - \sqrt{K^2 - 1}).$$

### Stability

- **Oscillation** is possible if  $|\Gamma_{in}| > 1$  or  $|\Gamma_{out}| > 1$  (negative resistance at a port).
- **Unconditionally stable**:  $|\Gamma_{in}|, |\Gamma_{out}| < 1$  for every passive source and load. Test (Rollett):

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} > 1 \text{ and } |\Delta| < 1$$

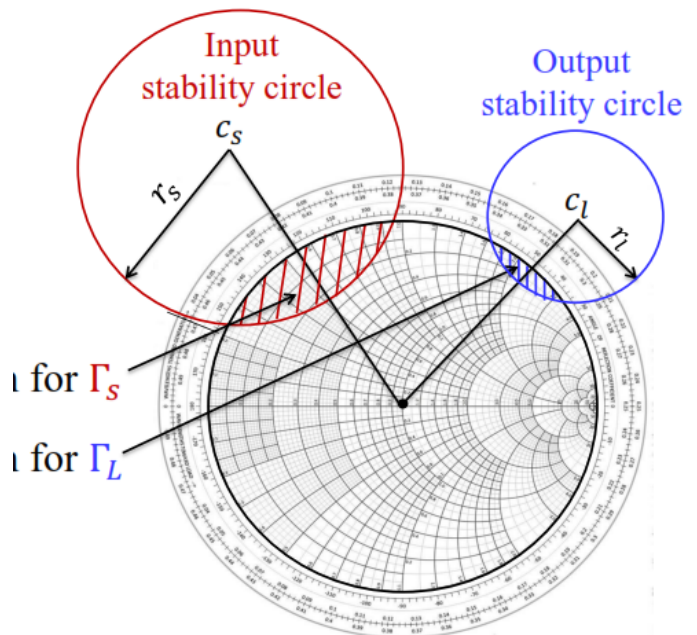
$$\text{or the single test } \mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|} > 1.$$

- **Conditionally (potentially) stable**: some  $\Gamma_S$  or  $\Gamma_L$  cause oscillation. Find them with **stability circles**:

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}, \quad R_L = \left| \frac{S_{12}S_{21}}{|S_{22}|^2 - |\Delta|^2} \right|$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}, \quad R_S = \left| \frac{S_{12}S_{21}}{|S_{11}|^2 - |\Delta|^2} \right|$$

- On the **output circle**  $|\Gamma_{in}| = 1$ ; on the **input circle**  $|\Gamma_{out}| = 1$ . Each is a boundary, not a region.



Deck slides 45 to 46:  $S_{11} = 0.65\angle -95^\circ$  etc. at 800 MHz,  $K = 0.547$ :

$C_S = 1.79\angle 122^\circ$ ,  $R_S = 1.04$ ;  $C_L = 1.30\angle 48^\circ$ ,  $R_L = 0.45$ . Hatched overlaps are the unstable  $\Gamma$ . Reproduced exactly in amp.py.

### Which side of the circle is stable (synthesis on the Smith chart)

1. Plot  $C_L$ ,  $R_L$  (and  $C_S$ ,  $R_S$ ) on the chart.
2. Test the **centre**  $\Gamma_L = 0$ : there  $|\Gamma_{in}| = |S_{11}|$ .
3. If  $|S_{11}| < 1$  the centre is stable, so the stable region is the side of the circle that contains the centre:
  - $|C_L| > R_L$  (centre outside)  $\Rightarrow$  stable **outside** the circle.
  - $|C_L| < R_L$  (centre inside)  $\Rightarrow$  stable **inside**.
4. Same for the source circle with  $|S_{22}|$ .
5. Circle entirely outside the unit chart (or enclosing it on the stable side)  $\Rightarrow$  unconditionally stable. Circle cutting the chart  $\Rightarrow$  conditionally stable; choose  $\Gamma_S$ ,  $\Gamma_L$  well inside the stable part.

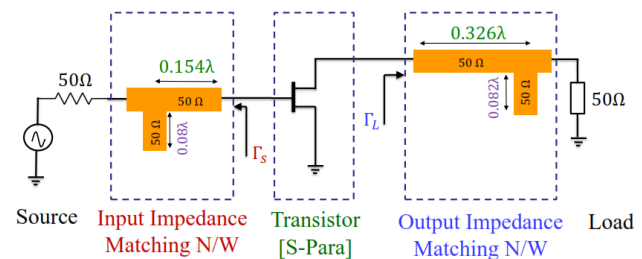
73 Bh, 74 Bh, 75 Bh give a sketched chart; 71 Bh gives the circles directly.

Worked: companion Problem 7.

### Amplifier design flowchart (write as boxes joined by arrows)

1. **Specify**  $f$ , gain, noise figure,  $Z_0$ ; **choose a transistor** (GaAs FET / HEMT / BJT) and bias; read its S-parameters.
2. **Stability**: compute  $\Delta$ ,  $K$  (or  $\mu$ ).
3.  $K > 1$ ,  $|\Delta| < 1$ ? **Yes**  $\rightarrow$  step 4. **No**  $\rightarrow$  draw stability circles, restrict  $\Gamma_S$ ,  $\Gamma_L$  to the stable region (or add resistive loading / feedback), then step 4.
4. **Unilateral check**: compute  $M$  and the error bound.
  - Small  $\rightarrow \Gamma_S = S_{11}^*$ ,  $\Gamma_L = S_{22}^*$  (or gain circles for a specified gain).
  - Large  $\rightarrow$  bilateral:  $\Gamma_S$ ,  $\Gamma_L$  from  $B$ ,  $C$ .
5. **Gain**:  $G_{TU,max}$  or  $G_{T,max}$ ; compare with the specification.
6. **Matching networks**: realise  $\Gamma_S$  and  $\Gamma_L$  from  $50\ \Omega$  with a line plus a stub (Smith chart, Ch 2).
7. **Bias network**, DC blocks, RF chokes; **layout** in microstrip; simulate, build, measure, tune.

Final Amplifier Circuit Diagram



Final amplifier: input network, transistor, output network. Deck slide 58.

### Amplifier circuit vs oscillator circuit in terms of stability factor 5

|                                    | Amplifier                                                                                        | Oscillator                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wanted                             | stable: no self-oscillation                                                                      | unstable: oscillation on purpose                                                                                                                            |
| Stability factor                   | $K > 1$ and $ \Delta  < 1$ (or $\mu > 1$ ), or $\Gamma_S$ , $\Gamma_L$ kept in the stable region | $K < 1$ : device potentially unstable                                                                                                                       |
| 71 Ma Port reflection Terminations | $ \Gamma_{in} ,  \Gamma_{out}  < 1$<br>conjugate match for gain                                  | $ \Gamma_{in}  > 1$ (negative resistance)<br>$\Gamma_S$ chosen <b>inside</b> the unstable region; $\Gamma_{in}\Gamma_S = 1$ ,<br>$\Gamma_{out}\Gamma_L = 1$ |
| Input signal                       | required                                                                                         | none; starts from noise                                                                                                                                     |
| Feedback                           | avoided, or negative                                                                             | positive, loop gain $A\beta \geq 1$                                                                                                                         |

### Stability and maximum gain numericals (bilateral and unilateral) 5+5 10 12 15 18

**TOP 22/24** 69 Bh, 70 Bh, 70 Ma, 71 Bh, 71 Ma, 72 Ash, 72 Ma, 73 Ma, 74 Bh, 74 Ma, 76 Bh, 77 Ch, 78 Ch, 79 Bh, 79 Ch, 80 Ba, 80 Bh, 80 Ch, 81 Ba, 81 Bh, 82 Ba, 82 Bh · **every set is solved in the companion**

- 13 distinct S-parameter sets across 22 papers; the companion works one in full, tabulates the rest with every intermediate, and splits out the cases that need a different answer (conditionally stable, given terminations, full matching design, circles given directly).

## 6.5 Microwave Oscillator Theory

### Microwave oscillator theory 8

PIN 2/24 70 Ma, 71 Ma · 71 Ma as the amplifier-vs-oscillator contrast of §6.4

#### Feedback view

- Gain  $A$  with positive feedback  $\beta$ :  $\frac{e_o}{e_i} = \frac{A}{1 - A\beta}$ .
- $A\beta = 1 \rightarrow$  finite output with no input: **Barkhausen condition**. To start,  $A\beta \approx 1.1$  to 1.2; oscillation grows from noise until gain compression brings  $A\beta$  back to 1.

#### Negative-resistance view (the microwave one)

- A device port with  $|\Gamma_{out}| > 1$  has **negative resistance**.
- One-port oscillator**: device  $Z_{out} = R_{out} + jX_{out}$  with  $R_{out} < 0$ , load  $Z_L$ . Loop condition  $\Gamma_{out}\Gamma_L = 1$  gives

$$R_L + R_{out} = 0, \quad X_L + X_{out} = 0$$

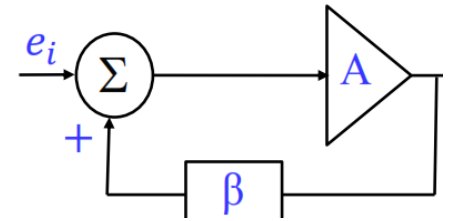
- To start oscillation the circuit must have net negative resistance:  $R_L < |R_{out}|$ . The deck's rule  $|R_{out}| \geq 1.2R_L$  (Pozar uses  $R_L = |R_{out}|/3$ ).
- Two-port oscillator**: transistor with a **generator tuning network** ( $\Gamma_S$ ) and a **load network** ( $\Gamma_L$ ). Conditions:
  - $K < 1$ : device potentially unstable.
  - $\Gamma_{in}\Gamma_S = 1$ .
  - $\Gamma_{out}\Gamma_L = 1$  (equivalent to 2, shown by substituting  $\Gamma_{in}$ ).

#### Two-port design steps

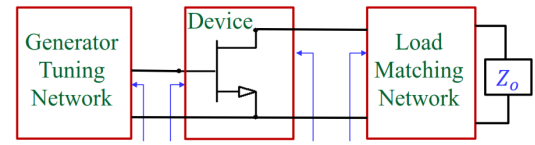
- From the S-parameters find  $K$ ; need  $K < 1$  (use a common-gate / common-base configuration or series feedback to make it so).
- Draw the **input stability circle**.
- Choose  $\Gamma_S$  **inside the unstable region**, usually on the rim for the largest  $|\Gamma_{out}|$ ; realise it with an inductor or shorted stub.
- $\Gamma_{out} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S}$ ,  $|\Gamma_{out}| > 1 \rightarrow Z_{out} = R_{out} + jX_{out}$  with  $R_{out} < 0$ .
- Choose  $R_L = |R_{out}|/3$  (start-up margin),  $X_L = -X_{out}$ ; design the load matching network.

#### Worked: Gunn diode one-port (deck slide 66, checked in amp.py)

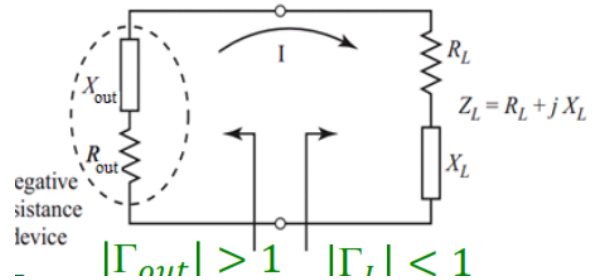
- $\Gamma_{out} = 1.24\angle 30^\circ$  at 10 GHz. Plot  $1/\Gamma_{out}^* = 0.806\angle 30^\circ$ , read  $z$  and negate the real part:  $Z_{out} = -68.9 + j159.0 \Omega$  (deck reads  $-70 + j160$ ).
- Resonance:  $X_L = -159 \Omega$ , a capacitor  $C = 1/(2\pi f \cdot 159) = 0.100$  pF.
- Start-up: **the deck picks  $R_L = 60 \Omega$ , which breaks its own rule**:  $|R_{out}| \geq 1.2R_L$  needs  $R_L \leq 57.5 \Omega$ . Use  $R_L = 23 \Omega$  (Pozar's  $|R_{out}|/3$ ) or at most  $57 \Omega$ . verified/added



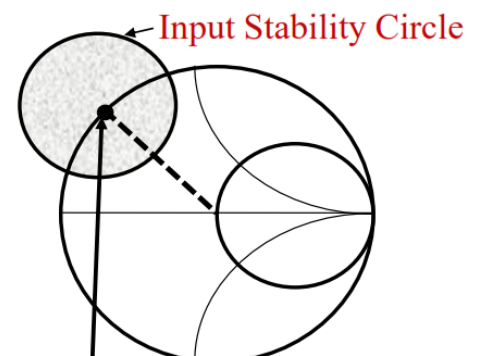
Positive feedback. Deck slide 60.



Two-port oscillator. Deck slide 62.



One-port: negative-resistance device and load. Deck slide 65.



$\Gamma_S$  chosen inside the input stability circle. Deck slide 67.

#### Oscillator types and stabilisation

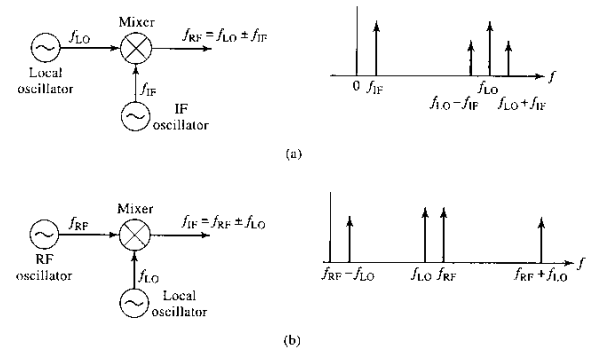
- Transistor (FET, BJT), Gunn, IMPATT (one-port); YIG-tuned (wide tuning), DRO (dielectric resonator, low phase noise).
- Frequency stability: high-Q resonator in the tuning network; isolator at the output (Ch 4).

## 6.6 Microwave Mixer

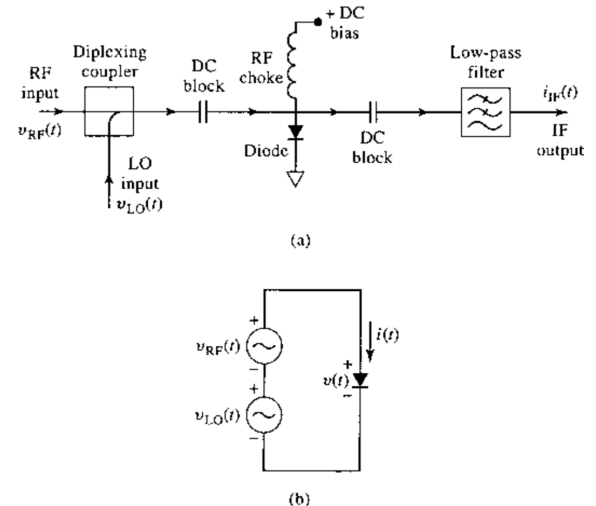
### Mixer theory / microwave mixer 5

PIN 2/24 70 Bh, 72 Ma · verified/added the deck has no mixer slides; written from Pozar Ch 13

- **Mixer:** a **nonlinear** three-port (diode or FET) that multiplies an RF signal  $f_{RF}$  with a local oscillator  $f_{LO}$  to translate it in frequency.
- **Square-law principle:**  $i = av^2$  with  $v = V_{RF} \cos \omega_{RF} t + V_{LO} \cos \omega_{LO} t$  contains  $aV_{RF}V_{LO} \cos(\omega_{RF} \pm \omega_{LO})t$ : **sum and difference** (checked by FFT in amp.py).
- **Down-conversion:** keep  $f_{IF} = |f_{RF} - f_{LO}|$  with a filter; the **superheterodyne** receiver then amplifies at a fixed IF.
- **Up-conversion** (transmitters): keep  $f_{RF} = f_{LO} + f_{IF}$ .
- **Image frequency:**  $f_{IM} = f_{LO} \pm f_{IF}$  on the other side of the LO also lands on  $f_{IF} \rightarrow$  reject it with an RF pre-filter or an image-reject mixer.
- **Figures of merit:** conversion loss  $L_c = 10 \log(P_{RF}/P_{IF})$  (typically 4 to 7 dB for diode mixers), noise figure, LO-RF and LO-IF isolation, 1 dB compression, intermodulation.
- **Types:**
  - **Single-ended diode:** one Schottky diode; simple, poor isolation.
  - **Balanced:** two diodes fed through a  $90^\circ$  or  $180^\circ$  hybrid (magic tee, rat race, Ch 4): cancels LO AM noise, better isolation, better input match.
  - **Double-balanced:** diode ring with two baluns: suppresses even harmonics and both RF and LO at the IF port.
  - **FET mixers:** conversion **gain** instead of loss.



(a) up-conversion, (b) down-conversion, each with its output spectrum: both  $f_{LO} \pm f_{IF}$  (or  $f_{RF} \pm f_{LO}$ ) appear and a filter keeps one. Pozar Fig 13.24.



(a) single-ended diode mixer: RF and LO combined in a diplexing coupler, DC-biased diode, low-pass filter to the IF port; (b) its ideal equivalent,  $v_{RF} + v_{LO}$  across the diode. Pozar Fig 13.25.

## 6.7 PYQ Mapping

### Theory questions, covered in this document

- **Insertion loss method; why it is used**  $\rightarrow$  §6.1. 72 Ash 5, 73 Bh 2+6, 73 Ma 8, 75 Bh 5, 80 Ba 3, 81 Bh 8, 82 Bh 3
- **Butterworth and Chebyshev common to prototype filters**  $\rightarrow$  §6.1. 70 Bh 4, 70 Ma 4
- **LPF prototyping (Butterworth, Chebyshev); filter parameters**  $\rightarrow$  §6.1. 72 Ash 5, 72 Ma 5, 74 Ma 3, 76 Bh 8, 77 Ch 8, 79 Bh 6, 80 Bh 10
- **Filter design steps; conversion to other filter types**  $\rightarrow$  §6.2. 71 Bh 5, 74 Bh 6, 80 Bh 10, 81 Ba 7, 82 Ba 7, 82 Bh 8
- **Microstrip implementation (LPF  $\pi$ , double  $\pi$ , double-pad T, shunt-arm, series-arm, HPF, BPF)**  $\rightarrow$  §6.3. 72 Ash 3, 73 Ma 2, 74 Bh 4, 74 Ma 5, 76 Bh 2, 78 Ch 10, 79 Bh 2, 80 Ba 5, 80 Bh 10, 81 Ba 3, 81 Bh 2, 82 Ba 3, 82 Bh 3
- **Design steps of a given 5th-order ladder (figure)**  $\rightarrow$  companion Problem 10. 82 Bh 3+8+3
- **Amplifier design flowchart**  $\rightarrow$  §6.4. 71 Bh 5, 73 Bh 10, 74 Bh 4, 76 Bh 4, 78 Ch 10, 81 Ba 3, 81 Bh 3, 82 Ba 4
- **Stability analysis; synthesis from a sketched Smith chart**  $\rightarrow$  §6.4 and companion Problem 7. 73 Bh 10, 74 Bh 8, 75 Bh 5, 81 Ba 4, 82 Ba 4
- **Stability of given circles  $C_S, R_S, C_L, R_L$**   $\rightarrow$  companion Problem 7. 71 Bh 5
- **Amplifier vs oscillator in terms of stability factor**  $\rightarrow$  §6.4. 71 Ma 5

- **Microwave oscillator theory** → §6.5. 70 Ma 8
- **Mixer theory** → §6.6. 70 Bh 5, 72 Ma 5

### Numerical content

- **This chapter has a numerical companion, Ch6 - RF Design Practices - Numerical Problems and Solutions:** every amplifier S-parameter set in the archive (22 papers) and the 82 Bh filter design. All values from `src/amp.py` and `src/filt.py`.

### Errors found and corrected

- Deck slide 51 (8 GHz example):  $|\Delta| = 0.168$ ,  $K = 3.53$ ; exactly 0.173 and 3.445 (verdict unchanged).
- Deck slide 52:  $M = 0.04$  and bounds  $-0.36$  to  $+0.37$  dB; exactly  $M = 0.034$ ,  $-0.29$  to  $+0.30$  dB.
- Deck slide 66:  $R_L = 60 \Omega$  violates the deck's own start-up rule.
- Deck example 8.6:  $N = 5$  already meets 20 dB;  $N = 6$  is conservative.
- Sorted PYQ documents: 71 Ma's set and marks, 70 Ma and 80 Ba asks, the missing 75 Bh Smith-chart question and 72 Ma mixer note (fixed before this chapter).

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short     | Means                                                       | Short    | Means                                                   |
|-----------|-------------------------------------------------------------|----------|---------------------------------------------------------|
| RF        | radio frequency                                             | MW, M/W  | microwave                                               |
| EM, EMR   | electromagnetic (radiation)                                 | TEM      | transverse electromagnetic mode                         |
| TE        | transverse electric mode                                    | TM       | transverse magnetic mode                                |
| WG        | waveguide                                                   | TL       | transmission line                                       |
| VSWR      | voltage standing wave ratio                                 | SWR      | standing wave ratio                                     |
| VNA       | vector network analyzer                                     | SA       | spectrum analyzer                                       |
| DUT       | device under test                                           | S-params | scattering parameters                                   |
| RL        | return loss                                                 | IL       | insertion loss                                          |
| ML        | mismatch loss                                               | NF       | noise figure                                            |
| BW        | bandwidth                                                   | SNR      | signal to noise ratio                                   |
| Tx, TX    | transmitter                                                 | Rx, RX   | receiver                                                |
| BTS       | base transceiver station                                    | GSM      | global system for mobile communications                 |
| EIRP      | effective isotropic radiated power                          | ERP      | effective radiated power                                |
| SAR       | specific absorption rate                                    | SARPs    | standards and recommended practices                     |
| HERP      | hazards of EM radiation to personnel                        | RADAR    | radio detection and ranging                             |
| ASR       | airport surveillance radar                                  | CW       | continuous wave                                         |
| Short     | Means                                                       | Short    | Means                                                   |
| TWT, TWTA | traveling wave tube (amplifier)                             | BWO      | backward wave oscillator                                |
| CFA       | cross-field amplifier                                       | LNA      | low noise amplifier                                     |
| HPA       | high power amplifier                                        | SWS      | slow wave structure                                     |
| O-type    | linear beam tube (axial $B$ -field)                         | M-type   | crossed-field tube ( $E \perp B$ )                      |
| MASER     | microwave amplification by stimulated emission of radiation | LASER    | light amplification by stimulated emission of radiation |
| MESFET    | metal-semiconductor FET                                     | HEMT     | high electron mobility transistor                       |
| GaAs      | gallium arsenide                                            | FET      | field effect transistor                                 |
| BJT       | bipolar junction transistor                                 | PIN      | p-type, intrinsic, n-type diode                         |
| IMPATT    | impact ionization avalanche transit-time                    | TRAPATT  | trapped plasma avalanche triggered transit              |
| LPF       | low pass filter                                             | HPF      | high pass filter                                        |
| BPF       | band pass filter                                            | BSF      | band stop filter                                        |
| MAG       | maximum available gain                                      | MSG      | maximum stable gain                                     |
| MIC       | microwave integrated circuit                                | MMIC     | monolithic microwave integrated circuit                 |
| ADS       | advanced design system                                      | IF       | intermediate frequency                                  |
| LO        | local oscillator                                            | RFIC     | radio frequency integrated circuit                      |
| PCB       | printed circuit board                                       | SMA      | subminiature version A connector                        |

| Short    | Means                | Short  | Means                            |
|----------|----------------------|--------|----------------------------------|
| attr(s)  | attribute(s)         | no.    | number                           |
| freq     | frequency            | info   | information                      |
| ops      | operations           | config | configuration                    |
| rm       | remove               | vs, Vs | versus                           |
| w/, w/o  | with, without        | Eg     | example / for example            |
| +Fig     | draw a figure        | (I)    | importance / need / significance |
| →        | leads to / therefore | +      | and                              |
| =        | is / equals          | norm   | normalized                       |
| max, min | maximum, minimum     | calc   | calculation / calculate          |
| eq, eqn  | equation             | approx | approximately                    |
| pts      | points               | coeff  | coefficient                      |
| temp     | temperature          | params | parameters                       |

**Standard and component terms** (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ( $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ ) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

**Key mathematical symbols:**  $Z_0$  characteristic impedance ·  $Y_0$  characteristic admittance ·  $\Gamma$  reflection coefficient ·  $S$  scattering matrix ·  $K$  Rollett's stability factor ·  $\Delta$  determinant of S-matrix ·  $\mu$  single parameter stability metric ·  $G_T$  transducer gain ·  $G_{TU}$  unilateral transducer gain ·  $f_c$  cutoff frequency ·  $\lambda_g$  guide wavelength ·  $v_p$  phase velocity ·  $v_g$  group velocity.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

**Tier chips** over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.